

(93)

Derivatives of inverse trigonometric functions

1. The function $y = \sin^{-1}x$ ($-1 < x < 1$) has its inverse $x = \sin y$ ($-\pi/2 < y < \pi/2$)

$$\Rightarrow y = \sin^{-1}x \text{ iff } x = \sin y.$$

$$\Rightarrow \frac{d}{dx} (\sin^{-1}x) = ?$$

Since $y = \sin^{-1}x \Rightarrow$ we differentiate with respect to x to obtain $\frac{dy}{dx}$

and $x = \sin y \Rightarrow$ we differentiate with respect to y to obtain $\frac{dx}{dy}$.

$$\Rightarrow \frac{dy}{dx} = \frac{1}{dx/dy}$$

$$\Rightarrow x = \sin y \text{ and } \frac{dx}{dy} = \cos y.$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\cos y}.$$

But the original question is in terms of x .

$$\text{Recall that } \sin^2 y + \cos^2 y = 1$$

$$= \cos^2 y = 1 - \sin^2 y.$$

$$\Rightarrow \cos y = \sqrt{1 - \sin^2 y}$$

(94)

$$\text{But } x = \sin y \Rightarrow \cos y = \sqrt{1-x^2}$$

$$\Rightarrow \frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$$

2. The function $y = \cos^{-1} x$ ($-1 < x < 1$) has its inverse $x = \cos y$ ($0 < y < \pi$) are inverses of each other.

$$\Rightarrow y = \cos^{-1} x \text{ iff } x = \cos y.$$

$$\Rightarrow \frac{d}{dx} (\cos^{-1} x) = \frac{1}{dx/dy} = \frac{-1}{\sin y}.$$

$$\sin^2 y + \cos^2 y = 1 \Rightarrow \sin y = \sqrt{1 - \cos^2 y} \\ = \sqrt{1 - x^2}$$

$$\Rightarrow \frac{d}{dx} \cos^{-1} x = \frac{-1}{\sqrt{1-x^2}}$$

3. The function $y = \tan^{-1} x$ ($-\infty < x < \infty$) has its inverse $x = \tan y$ ($-\pi/2 < y < \pi/2$)

$$\Rightarrow y = \tan^{-1} x \text{ iff } x = \tan y.$$

$$\Rightarrow \frac{d}{dx} (\tan^{-1} x) = \frac{1}{dx/dy} = \frac{1}{\sec^2 y}$$

$$\text{But } \sec^2 y = 1 + \tan^2 y = 1 + x^2.$$

$$\Rightarrow \frac{d}{dx} (\tan^{-1} x) = \frac{1}{1+x^2}.$$

(95)

4. The function $y = \cot^{-1} x$ ($-\infty < x < \infty$) has its inverse $x = \cot y$ ($0 < y < \pi$)

$$\Rightarrow y = \cot^{-1} x \text{ iff } x = \cot y.$$

Exercise: Show that

$$\frac{d}{dx} \cot^{-1} x = \frac{-1}{1+x^2}.$$

Example 1:

Find the derivative of the following inverse trigonometric functions

(a) $y = \tan^{-1}(2x)$.

Solution

$$y = \tan^{-1}(2x)$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{1+(2x)^2} \cdot 2 = \frac{2}{1+4x^2}.$$

(b) $y = \sin^{-1}\left(\frac{1}{x}\right)$

Solution

$$\text{If } y = \sin^{-1}\left(\frac{1}{x}\right) = \sin^{-1}(x^{-1})$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sqrt{1-\left(\frac{1}{x}\right)^2}} \cdot x^{-2}$$

$$= \frac{1}{\sqrt{\frac{x^2-1}{x^2}}} \cdot \frac{1}{x^2}$$

$$= \frac{-1}{x\sqrt{x^2-1}}.$$

(96)

(c) $y = \cos^{-1} x^2$
Solution

If $y = \cos^{-1} x \Rightarrow$

$$\frac{dy}{dx} = \frac{-1}{\sqrt{1-(x^2)^2}} \cdot 2x$$

$$= \frac{-2x}{\sqrt{1-x^4}}$$

(d) $y = \tan^{-1}(2x+1)$
Solution

If $y = \tan^{-1}(2x+1)$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{1+(2x+1)^2} \times 2 = \frac{2}{4x^2+4x+2}$$

$$= \frac{1}{2x^2+2x+1}$$

(e) $y = \cot^{-1}(3x^2)$

(f) $y = \cos^{-1}(4x)$

(g) $y = \tan^{-1}(\tan x)$

(h) $y = \sin^{-1}(\ln x)$

Logarithmic Differentiation

(97)

This is an application of implicit differentiation.

Example 1.

If $y = a^x$, find $\frac{dy}{dx}$.

Solution

If $y = a^x \Rightarrow$ Introducing $\ln$ on both sides we obtain

$$\ln y = \ln a^x = x \ln a.$$

$$\Rightarrow \ln y = x \ln a.$$

Differentiating on both sides with respect to x , we get

$$\frac{d}{dx}(\ln y) = \frac{d}{dx}(x \ln a)$$

$$\frac{1}{y} \frac{dy}{dx} = \ln a.$$

$$\Rightarrow \frac{dy}{dx} = y \ln a = a^x \ln a.$$

Example 2.

Differentiate $y = x^{\sin x}$

Solution

$$\text{If } y = x^{\sin x}$$

Introducing $\ln$ on both sides we obtain

$$\ln y = \ln x^{\sin x} = \sin x \ln x.$$

(98)

$$\ln y = \sin x \ln x$$

Differentiating on both sides with respect to x , we obtain

$$\frac{d}{dx} \ln y = \frac{d}{dx} (\sin x \ln x)$$

$$\frac{1}{y} \frac{dy}{dx} = \cos x \ln x + \sin x \cdot \left(\frac{1}{x}\right)$$

$$\Rightarrow \frac{dy}{dx} = y \left(\cos x \ln x + \sin x \left(\frac{1}{x}\right) \right)$$

$$= x^{\sin x} \left(\cos x \ln x + \sin x \left(\frac{1}{x}\right) \right)$$

Exercise:

Find the derivatives of the following functions

(a) $y = x^{x^x}$

(b) $y = (\tan x)^{\tan 2x}$

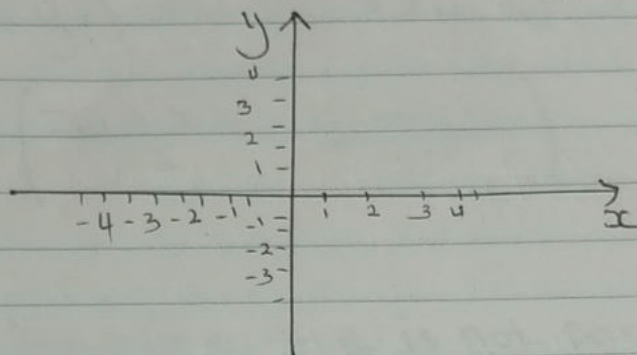
(c) $y = (1+x^2)^{\frac{1}{x^2}}$

(d) $y = x^x$

Analytical Geometry.

(99) (99)

Coordinate Systems in Two dimensions
i.e Cartesian Coordinate System.



● 1. Distance Formula.

The distance between any two points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ in a coordinate plane is given by.

$$d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Example:

● Prove that the triangle with vertices $A(-1, -3)$, $B(6, 1)$ and $C(2, -5)$ is a right triangle and find its area.

Solution

$$d(A, B) = \sqrt{(-1-6)^2 + (-3-1)^2} = \sqrt{65}$$

$$d(B, C) = \sqrt{(6-2)^2 + (1+5)^2} = \sqrt{52}$$

$$d(A, C) = \sqrt{(-1-2)^2 + (-3+5)^2} = \sqrt{13}$$

$$\Rightarrow [d(A, B)]^2 = [d(B, C)]^2 + [d(A, C)]^2$$

$$\text{Area} = \frac{1}{2} \times \sqrt{52} \times \sqrt{13} = 13 \text{ square units.}$$

2. Mid-point formula.

The mid-point of the line segment $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ is

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

3. lines:

Let L be a line that is not parallel to the y -axis, and let $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ be distinct points on L . The slope m (gradient) of L is given by

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x}.$$

NB: If L is parallel to the y axis, then the slope is not defined.

4. point-slope form

An equation for the line through the point $P(x_1, y_1)$ with slope m is

$$y - y_1 = m(x - x_1).$$

5. Slope-Intercept form.

The graph of the equation $y = mx + b$ is a line having slope m and y -intercept b .

Notice that every line is the graph of an equation of the form

$$ax + by + c = 0$$

where a , b and c are real numbers and a and b are not both zero.

Such an equation is called a linear equation, in x and y .

Theorem 1:

Two non vertical lines are parallel if they have the same slope.

Theorem 2:

Two lines with slopes m_1 and m_2 are perpendicular if $m_1 \times m_2 = -1$.

Exercise:

1. Find an equation for the perpendicular bisector of the line segment from $A(1, 7)$ to $B(-3, 2)$.

Ans: $8x + 10y - 37 = 0$.

2. Show that $A(-3, 1)$, $B(5, 3)$, $C(3, 0)$ and $D(-5, -2)$ are vertices of a parallelogram.
3. Prove that the points $A(6, 15)$, $B(11, 12)$, $C(-1, -8)$ and $D(-6, -5)$ are vertices of a rectangle.

4. If three consecutive vertices of a parallelogram are $A(-1, -3)$, $B(4, 2)$, $C(-7, 5)$, find the fourth vertex.
5. Find equations for the altitude of the triangle with vertices $A(-3, 2)$, $B(5, 4)$, $C(3, -8)$, and find the point at which they intersect.
6. Find a real number k such that the line $5x + ky - 3 = 0$ has y -intercept -5 .
7. Find a real number k such that the point $P(-1, 2)$ is on the line $kx + 2y - 7 = 0$.
8. Find all values of r such that the slope of the line through the points $(r, 4)$ and $(1, 3 - 2r)$ is less than 5.